Global–Local Transformer Network for HSI and LiDAR Data Joint Classification

Kexing Ding, Ting Lu, *Member, IEEE*, Wei Fu, Shutao Li, *Fellow, IEEE*, and Fuyan Ma

***Abstract*— Hyperspectral images (HSIs) contain rich spatial and spectral detail information, while light detection and ranging (LiDAR) data can provide the elevation information. Thus, the fusion of HSI and LiDAR data can help in more accurate image classification, which becomes a hot research topic. However, it is difficult to capture complex local and global spatial–spectral associations; meanwhile, how to build an effective interaction between multimodal data is another important issue. To this end, a novel global–local transformer network (GLT-Net) is proposed for the joint classification of HSI and LiDAR data, in this article. The main idea is to fully exploit the advantage of the convolution operator in characterizing locally correlated features and the promising capability of transformer architecture in learning long-range dependencies. Moreover, multiscale feature fusion and probabilistic decision fusion strategies are also designed in one framework, to further improve the classification performance. Here, the proposed GLT-Net mainly consists of multiscale local spatial feature learning, global spectral feature learning, and global–local feature fusion classification. In specific, multimodal image cubes of different sizes are first extracted and sent into convolutional neural networks (CNNs) to learn local spatial features, which is followed by multimodal information propagation and spatial-attention-guided multiscale feature fusion. Afterward, by considering spectral feature channels from a sequential perspective, vision transformers are introduced to model the global spectral dependencies. Finally, multiple class estimations based on local and global features are integrated via a probabilistic decision fusion strategy. In this way, complementary information of multimodal data and local/global spectral–spatial information can be fully mined and jointly used. Extensive experiments on three popular HSI and LiDAR datasets demonstrate that the proposed method performs superiority over the state-of-the-art methods. The source code of the proposed method will be made publicly available at https://github.com/Ding-Kexin/GLT-Net.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral images (HSIs), image classification, light detection and ranging (LiDAR) data, transformers.**

I. Introduction

HYPERSPECTRAL image (HSI) is a high-dimensional image with a large number of spectral bands, which can simultaneously reflect the spatial distribution information and detailed spectral reflection information of ground objects [1], [2], [3], [4]. Thus, the application of HSI for land cover classification has an important indicative role in many fields such as geological exploration, urban development planning, forest management, and precision agriculture [5], [6], [7], [8]. Light detection and ranging (LiDAR) data can not only record the elevation information of the observed objects but also have the advantages of less interference from environmental factors such as weather and occlusion [9]. Due to the differences in imaging mechanisms, HSI and LiDAR data can be used to obtain different aspects of ground objects, and their complementary information is beneficial to improve the performance of the land cover classification tasks [10], [11]. Therefore, it is a hot topic to study the image classification methods based on HSI and LiDAR information fusion.

In the past decades, a large number of HSI and LiDAR fusion classification methods have emerged, focusing on how to design effective feature extraction operators and appropriate multimodal information fusion classification strategies [12], [13], [14], [15], [16]. These methods are accustomed to exploit the inherent properties of the data for hand-designed feature fusion, such as raw spectral features and morphological filtering-based features. For example, in [15], features extracted from HSI and LiDAR data based on the morphologically extended attribute profiles are superimposed, enabling fusion of spectral, spatial, and elevation features for better classification. Liao et al. [17] combine the original hyperspectral spectral features and the dimensionality-reduced morphological profile features from HSI and LiDAR data through a graph-based generalized feature fusion method for the joint HSI and LiDAR classification. In addition, in some works, the joint use of feature-level fusion and decision-level fusion has been shown to further improve the classification performance. For example, in [18], the Gabor features extracted from HSI and LiDAR data and their amplitude and phase features are concatenated and fed into the classifiers, respectively. The three classification results are then regularized by two superpixel segmentation algorithms. Finally, the decision fusion strategy of weighted majority voting can effectively improve the utilization efficiency of multifeatures. However, the above-mentioned early works mainly rely on artificially designed features, which contain too much subjective ideas and cannot adaptively generalize the intrinsic features of multimodal data.

Recently, deep learning (DL) networks have achieved many outstanding results in the field of multimodal image fusion and analysis [19], [20], [21]. Convolutional neural networks (CNNs) in the DL methods have powerful feature extraction capabilities and can automatically learn deep semantic

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Kexing Ding, Ting Lu, Shutao Li, and Fuyan Ma are with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: ting_lu@hnu.edu.cn).

Wei Fu is with the College of Information Science and Engineering, Hunan University, Changsha 410082, China (e-mail: fuwei@hnu.edu.cn).

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features from images. Therefore, some feature-level fusion approaches of HSI and LiDAR data based on CNN depth features have emerged [22], [23], [24], [25]. In [11], the extension profile features extracted from HSI and LiDAR are fed into CNNs for further deep semantic feature extraction. This is the first attempt to fuse HSI and LiDAR features and classify using a DL model. In [26], a single-stream DL architecture based on dynamic grouped convolution is used for HSI and LiDAR joint classification. Xu et al. [27] use two independent CNN branches to extract deep features for HSI and LiDAR, respectively, and then the extracted features are concatenated and fused for classification. Later, some advanced spectral–spatial CNN-based methods for HSI classification have been extensively studied. Gao et al. [28] propose a multiscale residual network to learn more discriminative spectral–spatial feature representation. In [29], an attention-based adaptive spectral–spatial kernel improved residual network is designed, which exploits spectral attention to capture discriminative spectral–spatial features. Gradually, the spectral–spatial CNN-based methods are used for joint HSI and LiDAR classification. In [30], the spatial representation of HSI and the spectral representation of LiDAR data are further enhanced in a DL network, enabling sufficient interaction of cross-modal information. In [31], spectral–spatial features from HSI obtained by CNN are fused with elevation features from LiDAR obtained by spatial morphological blocks for efficient multimodal feature learning. Compared with the traditional artificially designed feature-level fusion methods, the CNN-based deep feature fusion approaches can automatically and adaptively learn representative features, which have great potential to improve the multimodal image classification task. However, although the existing CNN-based HSI and LiDAR joint classification methods perform well in extracting local spatial features, they cannot fully exploit the global sequence properties of spectral features.

More lately, transformer networks have been introduced into the field of computer vision (CV) due to the unique and powerful global modeling capabilities and achieved excellent results. In [32], transformer networks are directly applied to image patch sequences and perform well on the image classification tasks. Furthermore, some scholars apply transformers to the HSI classification tasks. In [33], convolutional and transformer structures are combined to capture the global–local features of HSI patches for better classification performance. In [34], transformer structures are adopted to mine and represent the local sequence properties of spectral features, and skip connections are designed to further improve the classification results. Thanks to the powerful global modeling ability and unique self-attention mechanism of transformers, the long-range feature extraction ability and distinguishability performance of the model are effectively improved. Therefore, it is also a promising direction to explore transformers to process and analyze multimodal data corresponding to the same objects from a global perspective. For example, in [35], convolution-guided transformers are used to learn the intrinsic relationship between local and global features for better fusion of visible and infrared images. However, to the best of our knowledge, it is rare to explore the application of transformers in HSI and LiDAR data fusion classification.

Based on the above analysis, this article attempts to propose a novel end-to-end DL framework, called the global–local transformer network (GLT-Net), for more accurate HSI and LiDAR data joint classification. The main motivation of the proposed GLT-Net is to take full advantage of the powerful local spatial feature extraction capability of CNN and the advantages of the transformer architecture in learning global spectral context information. On one hand, to extract better local features, local spatial feature learning for multimodal multiscale input is performed based on CNN. Differently, multiscale feature extraction, linear weighting fusion, and spatial-attention-based grouped dimensionality reduction (SA-GDR) module are designed to improve the local feature learning capability of CNN. On the other hand, the multihead attention mechanism of the transformer structure is applied in the spectral dimension, so as to realize the learning of global spectral sequence properties. Specifically, the feature learning process adopts an encoder–decoder structure, and a designed reconstruction loss is added to constrain the entire feature learning process. Then, the global–local prediction information is integrated by means of decision fusion for more accurate classification. The feature learning process and the fusion classification process are cotrained in a framework to make the model more robust and discriminative by optimizing a defined joint loss function consisting of a reconstruction loss and a supervised classification loss. Extensive experiments on three popular HSI and LiDAR datasets show that the proposed method outperforms other state-of-the-art methods both objectively and subjectively. The main contributions of this article are summarized as follows.

1) In this article, the multiscale aggregated CNN is used for local spatial feature extraction of HSI and LiDAR data. A multimodal information integration and an SA-GDR module for multiscale information fusion are designed, which can achieve effective interaction of multimodal and multiscale local features while improving the computational efficiency.
2) To fully extract and use the global spectral sequence properties, the transformer network is introduced for the first time for the fusion classification of HSI and LiDAR data. By considering spectral feature channels from a sequence perspective, GLT-Net is able to learn global spectral dependencies from grouped local spatial embeddings using a multihead attention mechanism.
3) To generate more accurate and robust classification results, a global–local fusion classification strategy is proposed. Specifically, the final classification result is obtained by the fusion of adaptive weighted decision-making based on the class probability estimation of local spatial features and global spectral features. We also verified the effectiveness of the proposed method through extensive experiments.

The rest of this article is organized as follows. In Section II, we describe the details of our proposed network. Section III presents the experimental data, results, and analysis of different models. Finally, the conclusion is drawn in Section IV.

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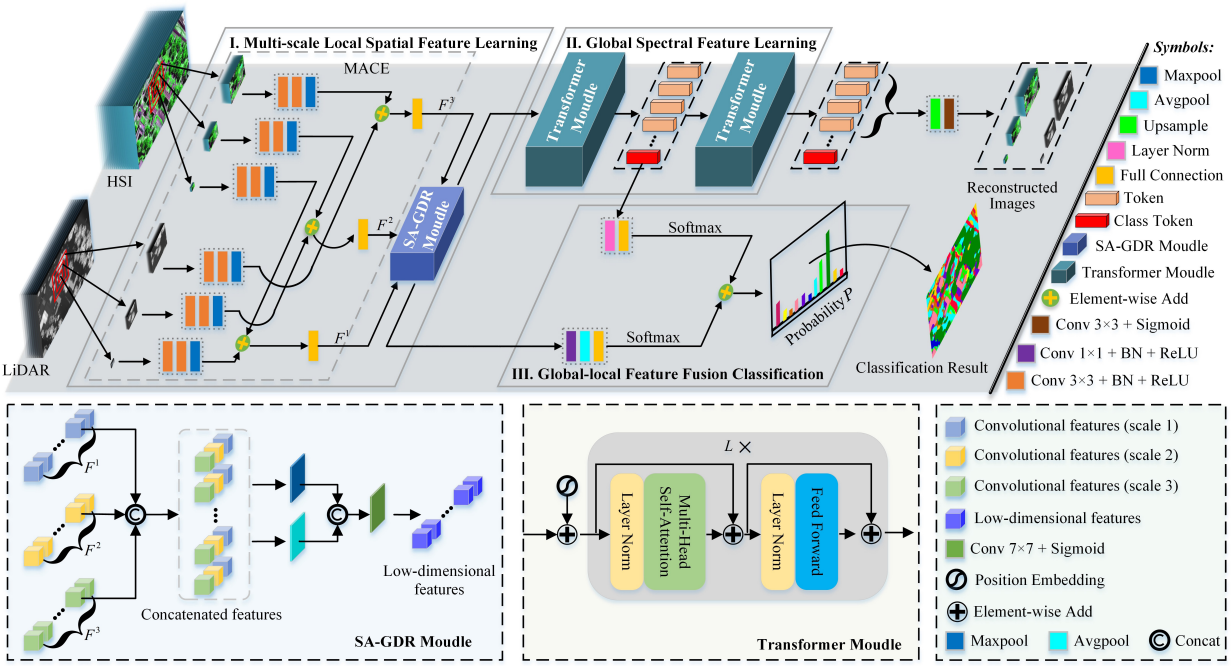


Fig. 1. Proposed framework of the GLT-Net method for HSI and LiDAR data joint classification.

II. Proposed Method

Benefiting from the powerful local feature extraction ability of CNN and the global context modeling ability of the transformer model, we try to combine them organically and apply them to the joint classification task of HSI and LiDAR. By proposing a GLT-Net, the multiscale local spatial neighborhood features and global spectral sequence features are fused, while the global–local multimodal complementary information is used by the classifier to improve the classification ability. As shown in Fig. 1, the proposed GLT-Net method mainly consists of three parts: multiscale local spatial feature learning (MLSFL, Part I), global spectral feature learning (GSFL, Part II), and global–local feature fusion classification (GLFFC, Part III). MLSFL adopts multiscale aggregation CNNs and aims to enhance the learning ability of local spatial features through multimodal multiscale feature fusion. GSFL applies a multihead attention mechanism in the spectral dimension to learn global spectral sequence properties. The purpose of GLFFC is to further strengthen multimodal complementary information fusion by combining local spatial convolutional encoding features with global spectral sequence encoding features, thereby promoting the improvement of classification results. The unsupervised multiscale reconstruction task and the supervised classification task are constrained by a designed overall loss function. Below, we will demonstrate the details of MLSFL, GSFL, and GLFFC, as well as the designed overall loss function.

A. Multiscale Local Spatial Feature Learning

The specific structure of MLSFL is shown in Part I of Fig. 1. Generally speaking, classical transformers are insufficient in multiscale representation ability and local spatial feature extraction. To this end, we design multiscale aggregated CNNs for multiscale local spatial feature extraction and fusion of HSI and LiDAR data. MLSFL consists of a multiscale aggregated convolutional encoder (MACE) and an SA-GDR module. Briefly, an MACE extracts multiscale local spatial features and performs adaptive linear weighted fusion of multimodal features. The SA-GDR module performs multiscale feature integration for the spectral properties of hyperspectral data. On one hand, multimodal and multiscale local spatial features are fully exploited. On the other hand, the dimension of fusion features is reduced, which is beneficial to improve the computational efficiency and subsequent global spectral feature extraction. Their specific descriptions are as follows:

1) Multiscale Aggregated Convolutional Encoder: The network composition of the MACE can be directly seen in Part I of Fig. 1. MACE includes a convolutional encoding part, an adaptive linear weighting operation, and a full connection layer. First, we set the base size parameter λ of the input data. As can be seen from the figure, the inputs to MACE are cubes around the HSI target pixels and the LiDAR target pixels, and their spatial sizes are $\lambda \times \lambda$, $2\lambda \times 2\lambda$, and $3\lambda \times 3\lambda$, respectively.

The encoding part is divided into three blocks, the first two blocks in turn contain 2-D convolution with kernel size 3×3, batch normalization, and rectified linear unit (ReLU) activation operations, and the number of convolution kernels is empirically set to 32 and 64. To reduce the amount of computation and avoid the problem of overfitting, we choose ReLU as the activation function here, which is also adopted in [36], [37], [38], [39], [40]. More importantly, we choose the above parameters to compare as fairly as possible with other classification methods. Notably, each convolutional layer uses a padding operator to keep the output spatial size constant. The third block is the max-pooling operation, and the size

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of the pooling window is 2×2. Each HSI cube $X_h^i \in \mathbb{R}^{w^i \times w^i \times d_h}$ and LiDAR cube $X_l^i \in \mathbb{R}^{w^i \times w^i \times d_l}$ are mapped to the same spectral dimension (32) by the first block, where $w^i = i \times \lambda, i \in \{1, 2, 3\}$ is the space size of the cubes, and d_h and d_l are the original spectral dimensions of HSI and LiDAR, respectively. After the second encoding block, the multiscale local spatial features are further extracted, and the cubes' number of spectral channels becomes 64. Next, to increase the receptive field and avoid feature invariance of the convolution layer, the max-pooling operation in the third encoding block reduces the size of the cubes by half.

Assuming that the multiscale HSI and LiDAR features obtained through the above encoding part are F_h^i and F_l^i, respectively, then they are adaptively linearly weighted to achieve multimodal feature fusion. The fusion feature f^i of the ith scale can be calculated according to the following formula:

$$f^i = \alpha F_h^i + (1 - \alpha) F_l^i \tag{1}$$

where α is an adaptive weighting coefficient, which can be automatically learned and updated by the network. Due to the strong difference between HSI and LiDAR data, the initial value of α is set to 0.5 to ensure that F_h^i and F_l^i have equal importance.

Finally, a fully connected layer maps the fused features to the same spatial size for subsequent feature extraction, and we denote them as F^i. In general, through MACE, on one hand, local spatial features are fully extracted, and on the other hand, multimodal features at the same scale are effectively fused.

2) SA-GDR Module: The detailed structure of the SA-GDR module can be clearly seen in Fig. 1, with F^i as inputs. Inspired by the idea in [41], the spectral reflectance information of adjacent bands in HSI tends to be similar. Since the convolution operation does not break the order of the spectral dimensions, we believe that the fused features F^i also have the same property. In SA-GDR, F^i is first divided into k groups in channel order. Then the jth group of fusion features $F^i(j)$ corresponding to the input of three scales is first concatenated along the spectral dimension to obtain the concatenated feature F_c^j, as shown in the following formula:

$$F_c^j = \underset{i \in \{1,2,3\}}{\text{Concat}} \left(F^i(j) \right), \quad j \in \{1, 2, \ldots, k\}. \tag{2}$$

Next, global average pooling and global max-pooling are performed for the channel dimension of F_c^j to generate two feature maps F_m^j and F_a^j representing different processing methods and attributes. In this way, the spatial features are further emphasized by the attention mechanism. The processing procedure can be expressed as

$$F_m^j = \text{Maxpooling} \left(F_c^j \right) \tag{3}$$

$$F_a^j = \text{Avgpooling} \left(F_c^j \right). \tag{4}$$

Then F_m^j and F_a^j are combined into 2-D features. Finally, we use 7×7 convolution with larger receptive field and sigmoid operation for feature fusion and logistic regression to get a low-dimensional feature F^j. The processing procedure can be written as follows:

$$F^j = \text{Sigmoid} \left(\text{Conv} \left(\text{Concat} \left(F_m^j, F_a^j \right) \right) \right). \tag{5}$$

In this way, the feature maps of three k groups based on three scales are fused and connected to obtain a final low-dimensional local spatial feature F_{local} with dimension k. The processing procedure can be expressed as

$$F_{\text{local}} = \underset{j \in \{1,2,\ldots,k\}}{\text{Concat}} \left(F^j \right). \tag{6}$$

In general, after the SA-GDR module, on one hand, the dimension of fusion features based on multiscale input is reduced, which reduces the network overhead. On the other hand, multiscale local spatial features are effectively fused according to spectral characteristics.

B. Global Spectral Feature Learning

In previous works, the global spectral information in the classification task of HSIs and LiDAR data has rarely been paid attention to. Actually, different bands reflect various spectral reflection information of the same ground objects, and thus, they will provide different and complementary information for class identification. For example, some bands are more sensitive to identify several classes, while other bands are more advantageous to separate other classes. Therefore, how to effectively extract the complex spectral correlation between different bands in a global view will help in the improvement of classification results. Due to powerful global modeling capabilities of vision transformer (ViT), it is very suitable for studying the relationships between long sequence data. Based on the above motivations, we adopt the classical ViT structure to construct the correlation of global spectral information. Specifically, feature maps are mapped to different subspaces along spectral channels to learn global spectral dependencies.

The specific composition of GSFL is shown in Part II of Fig. 1. GSFL adopts an encoder–decoder structure as a whole, and both the encoder and the decoder use the classic ViT framework in [32]. A ViT encoder is used to further explore global spectral dependencies from convolutional encoding features F_{local} containing rich local spatial information. Inspired by the idea in [42], the reconstruction task has a good auxiliary role for downstream tasks. The purpose of the ViT decoder is to recover the spectral details of the original image, assisting the learning of spectral sequence information.

The composition of the transformer module is shown in Fig. 1, and we set the depth L of the ViT encoder to 5 and the encoded embedding dimension to 64. To consider the spatial location information, the learnable position encoding is first superimposed to F_{local}. Assuming that the feature obtained after superimposing the position information is denoted as x_0, the process of going through the ViT encoder can be described as

$$\hat{x}_l = x_{l-1} + \text{MSA}\left(\text{LN}\left(x_{l-1}\right)\right), \quad l \in \{1, 2, \ldots, L\} \tag{7}$$

$$F_{\text{global}} = \hat{x}_L + \text{FFN}\left(\text{LN}\left(\hat{x}_L\right)\right) \tag{8}$$

where y is the output of the ViT encoder, and FFN, LN, and multihead self-attention (MSA) denote the feedforward networks, layer normalization, and MSA, respectively. According to (7) and (8), the feature map F_{global} reflecting the properties

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of the global spectral sequence is obtained after L transformer blocks.

Notably, we decided to apply a multihead attention mechanism in the spectral dimension due to the learning of global spectral dependencies without destroying local spatial information. Specifically, first convert the input feature vector x_l into query, key, and value (i.e., Q_l, K_l, and V_l, respectively), then map the 3-D Q_l, K_l, and V_l matrix to h latent spaces, and finally calculate the attention value. The calculation process of the global spectral dependency based on the MSA mechanism is as follows:

$$\mathrm{MSA}\left(Q_l, K_l, V_l\right) = \mathrm{Softmax}\left(\frac{Q_l K_l^T}{\sqrt{d/h}}\right) V_l \tag{9}$$

where $l \in \{1, 2, \ldots, L\}$, Softmax is the softmax function, T represents the transpose operation, h is the number of heads, and d is the spectral dimension of the input feature vector x_l (we set it to be the same as the encoded embedding dimension, i.e., 64).

Corresponding to the ViT encoder, the ViT decoder is also divided into L blocks, and the decoded embedding dimension to 32. We also use a learnable position encoding and multihead attention mechanism for the decoder, with the same number of heads as the encoder.

As a summary, we adopt a vision ViT encoder and a ViT decoder to achieve learning and recovery of global spectral sequence relationships, respectively. Furthermore, as shown in Fig. 1, the class token F_{cls} obtained by the ViT encoder is used for the subsequent classification task, and the token F_{rec} obtained by the ViT decoder is used for the auxiliary reconstruction task. Such a multitask mode can enhance the feature extraction capability and the robustness of the network.

C. Global–Local Feature Fusion Classification

Part III of Fig. 1 shows the specific composition of GLFFC. To fully use the local spatial information and global spectral information of HSI and LiDAR, we introduce a decision fusion classification strategy. GLFFC is mainly divided into three steps: class prediction for F_{local}, class prediction for F_{cls} from F_{global}, and adaptive decision fusion. Suppose M is the number of samples in each batch.

First, F_{local} is classified by a CNN-based classifier, which contains three blocks. The first block sequentially contains 2-D convolution with kernel size 1×1, batch normalization, and ReLU activation operations, and the number of convolution kernel is set to 32. The second block is an adaptive global average pooling operation, which aims to reduce the space size to 1×1 to reflect the pixel category information. The third block is a fully connected layer, and the number of spectral channels is mapped to the set number of ground object categories C. Then through the softmax function, the C-dimensional features are processed into C-dimensional 0–1 probabilities P_{local}, which reflect the object category of the target pixel.

Second, $F_{\mathrm{cls}} \in \mathbb{R}^{M \times 1 \times d}$ is classified by a multilayer perceptron (MLP), which contains two blocks. The first block is the layer normalization operation, which is more suitable for spectral sequences with variable length than batch normalization. The second block is a fully connected layer, and the number of spectral channels is mapped to C as well. Then similar to the former, the softmax function processes the C-dimensional features into C-dimensional 0–1 probabilities P_{global}.

Finally, P_{local} and P_{global} corresponding to the encoded features of CNN and transformer are fused by adaptive decision, which can realize the organic combination of local spatial features and global spectral features and obtain better classification results. The final fusion classification probability P can be calculated as

$$P = \beta P_{\mathrm{local}} + (1 - \beta) P_{\mathrm{global}} \tag{10}$$

where β is a weighting coefficient that can be automatically learned and updated by the network to balance the importance of P_{local} and P_{global}. Since both the local spatial information and global spectral information are important, the initial value of β is set to 0.5 to ensure that P_{local} and P_{global} have equal importance.

To sum up, if it is assumed that the fusion classification probability corresponding to the sample obtained by the mth target pixel in a batch is P_m, the final classification result R_m can be expressed as

$$R_m = \underset{c \in [1, C]}{\arg\max} \; P_m(c), \quad \text{s.t.}, m \in [1, M]\,. \tag{11}$$

D. Overall Loss Function

In the proposed GLT-Net method, the feature learning process in MLSFL and GSFL is constrained by the reconstruction task, while GLFFC is used for the supervised joint classification task with label information. The cotraining of these two tasks makes the model more discriminative and predictive. According to the specific task requirements, we design a suitable overall loss function, which consists of two parts: reconstruction loss and classification loss. These loss functions are discussed in detail below.

1) Reconstruction Loss: As shown in Fig. 1, the token F_{rec} obtained by the ViT decoder in GSFL is used to assist the reconstruction task, which has captured local spatial features and global spectral dependencies. After upsampling with different scales, F_{rec} is then reconstructed through the 3×3 convolution and the sigmoid function operators as cube $Y_h^i \in \mathbb{R}^{w^i \times w^i \times d_h}$ and cube $Y_l^i \in \mathbb{R}^{w^i \times w^i \times d_l}$, corresponding to the input HSI cube $X_h^i \in \mathbb{R}^{w^i \times w^i \times d_h}$ and LiDAR cube $X_l^i \in \mathbb{R}^{w^i \times w^i \times d_l}$. The reconstruction loss $\mathcal{L}_{\mathrm{rec}}$ of each sample is computed based on the mean squared loss function, which can be modeled as follows:

$$\mathcal{L}_{\mathrm{rec}} = \frac{1}{M} \sum_{i=1}^{3} \sum_{m=1}^{M} \left(X_h^{im} - Y_h^{im}\right)^2 + \left(X_l^{im} - Y_l^{im}\right)^2. \tag{12}$$

L_{rec} makes the reconstruction cubes as similar to the input cubes as possible, constraining the entire feature learning process. The reconstruction branch improves the classification performance by enhancing the discriminative feature extraction ability.

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2) Classification Loss: The classification loss $\mathcal{L}_{\text{cls}}$ is calculated based on the cross-entropy loss function and can be modeled as follows:

$$\mathcal{L}_{\text{cls}} = \frac{1}{M}\sum_{m=1}^{M} y_m \log(P_m) + (1 - y_m)\log(1 - P_m) \quad (13)$$

where M is the number of samples in each batch, P_m is the predicted class probability corresponding to the mth sample calculated according to (10), and y_i represents the class label of the mth input cube in a batch. The classification accuracy is improved under the constraints of $\mathcal{L}_{\text{cls}}$.

3) Overall Loss: The final overall loss function $\mathcal{L}$ is composed of linear weighted reconstruction loss and classification loss, expressed as follows:

$$\mathcal{L} = \gamma\,\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{cls}} \quad (14)$$

where γ is the weight coefficient. It can be seen from Fig. 1 and (12) that there are six reconstruction targets designed in our method, which are HSI cubes and LiDAR cubes of three scales. Whereas in the classification task, we normalize only one fused feature. To balance the importance of the reconstruction task and the classification task as much as possible, we empirically set the weight coefficient of the reconstruction task to 1/6.

In summary, the overall loss function facilitates both the reconstruction and classification tasks, enabling global–local feature extraction and fusion classification within an end-to-end framework.

III. Experiments and Analysis

In this section, we conduct experiments based on three widely adopted HSI and LiDAR datasets to evaluate the capability of the proposed GLT-Net method and compare it with other advanced models.

A. Data Description

1) Houston2013 Data: The first scene was collected by the National Center for Airborne Laser Mapping (NCALM) in June 2012 using the Compact Airborne Spectrographic Imager (CASI) sensor over the University of Houston campus and adjacent urban areas. In particular, this dataset is published by the IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Competition. It consists of an HSI and a LiDAR-based digital surface model (DSM), both with a data size of 349 × 1905 pixels and a spatial resolution of 2.5 m. The HSI contains 144 bands with wavelengths ranging from 0.38 to 1.05 μm. The dataset has a total of 15 029 ground-truth samples covering 15 classes [43]. The data and ground-truth map provided by the IEEE GRSS Data Fusion Competition can be obtained from the IEEE GRSS data and algorithm standard evaluation (DASE) (http://dase.grss-ieee.org/). Table I lists the number of training and testing samples we used.

2) Trento Data: The second scene was captured in a rural area in southern Trento, Italy. The dataset contains HSI and one LiDAR data with a size of 166 × 600 pixels and a spatial resolution of 1 m. The HSI is collected by an airborne hyperspectral imaging systems (AISA) Eagle sensor with 63 spectral bands ranging from 0.42 to 0.99 μm in wavelength. The corresponding LiDAR data were acquired by an Optech Airborne Laser Topographic Mapping (ALTM) 3100EA sensor. The dataset contains a total of 30 214 ground-truth samples, and six distinguishable class labels are investigated [10]. Table II shows the training and testing samples.

3) MUUFL Data: The third scene was collected in November 2010 at the University of Southern Mississippi's Gulfport campus in Long Beach, Mississippi, USA. The dataset contains on-campus HSI-based and LiDAR-based DSMs with a size of 325 × 220 pixels. The HSI was collected by the ITRES Research Limited (ITRES) Compact Airborne Spectral Imager (CASI-1500) sensor with 64 available spectral channels ranging from 0.38 to 1.05 μm with a spatial resolution of 0.54 × 1.0 m. LiDAR data are captured by an ALTM sensor using a laser at 1064-nm wavelength. The spatial resolution is 0.60 × 0.78 m. The dataset contains a total of 53 687 ground-truth samples, and 11 distinguishable class labels are studied [23]. Table III lists the training and testing samples.

B. Experimental Setting

In our experiments, the proposed GLT-Net method is implemented in the PyTorch framework. The experiments were performed on a server with Intel(R) Xeon(R) Silver 4216 CPU at 2.10 GHz with 24G video random-access memory (VRAM). We use an Adam optimizer with a learning rate decay parameter of 0.9, and the batch size, the initial learning rate, and the number of training sessions are set to 64, 0.0005, and 500, respectively. We empirically set the batch size to 64 because the same settings were used in most of the comparative experiments and achieved excellent performance. The reason why the initial learning rate is set to 0.0005 is that we initially wanted to set a small learning rate to make the network optimization step smaller, so as to achieve more accurate feature learning. We combined our observations on the experimental process (test accuracy, etc.) and finally set the epoch to 500.

To objectively reflect the classification performance, different evaluation metrics are used, including the accuracy of each category, overall accuracy (OA), average accuracy (AA) and Kappa coefficient. For a fairer and more robust comparison, all the evaluation metrics take the average of ten classification results.

C. Parameter Discussion

The size of the input data has a certain influence on the extraction of local spatial information, and the performance of the transformer model is often related to the amount of data. Based on the above considerations, we design experiments to discuss the effect of the cubes' spatial size. In MLSFL, we set the basic spatial dimension parameter λ, so that the spatial sizes of the three input data are $\lambda \times \lambda$, $2\lambda \times 2\lambda$, and $3\lambda \times 3\lambda$, respectively. When λ takes a higher value, it means that the input is more informative, but this is affected by the spatial resolution of the dataset and the complexity of the scene. In general, lower spatial resolution and higher scene complexity result in more structural information in the neighborhood, and correspondingly larger coverage requirements [44]. As shown in Fig. 2, the effects of λ show similar trends on three different datasets, that is, too small or too large

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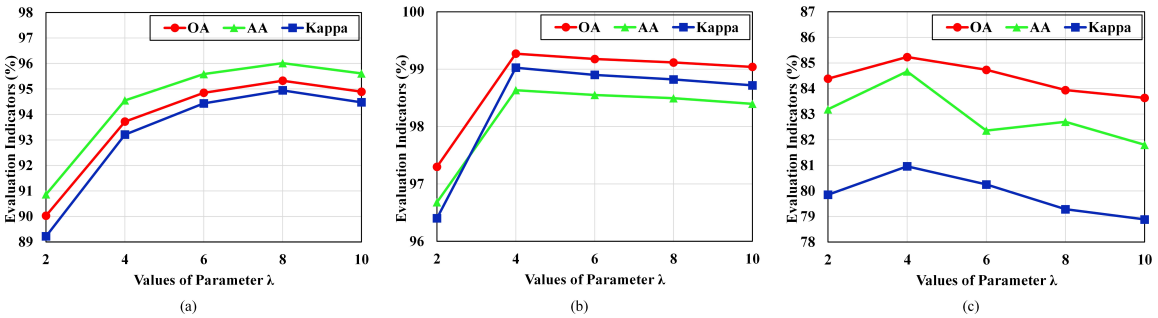


Fig. 2. Effects of the base size parameter λ on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL.

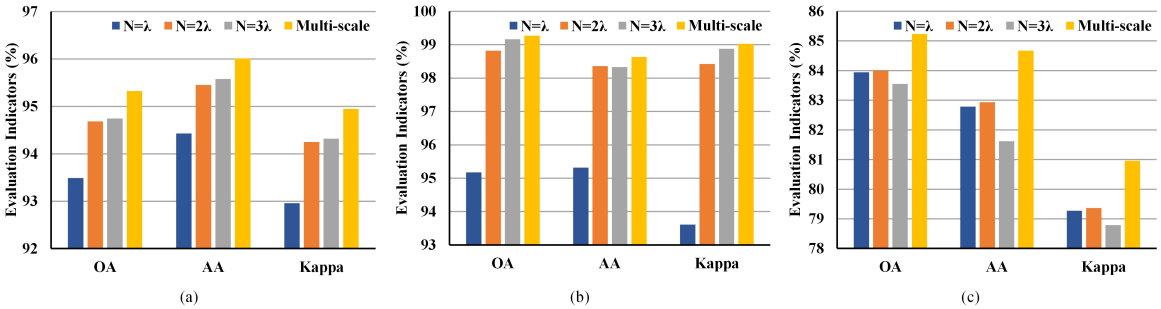


Fig. 3. Effects of multiscale inputs on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL.

λ values will lead to a decline in three objective indicators. Taking the MUUFL Gulfport (MUUFL) dataset as an example, a smaller value of λ (e.g., $\lambda = 2$) indicates that the space size of the input data is small, resulting in less information that the model can refer to each time, which limits the classification performance. In contrast, when λ is set too large (e.g., $\lambda = 10$), the input data exceed the coverage requirement due to its high spatial resolution and low scene complexity. Overall, it is critical to choose an appropriate input size for the GLT-Net method. Specifically, for the Houston2013 data, a relatively large input achieves the best results when the underlying input size is 8×8 (i.e., $\lambda = 8$). While for the Trento and MUUFL data, a relatively small base input size of 4×4 (i.e., $\lambda = 4$) provides the best performance. The above phenomenon is completely reasonable, because the Houston2013 data have lower spatial resolution and higher scene complexity. Similarly, the experimental results based on the Trento and MUUFL data are also completely convincing.

D. Ablation Studies

1) Effect of Multiscale Inputs: As shown in (1) and (2), we successively process multimodal information of the same scale and fused features of different scales in MLSFL, which can fully explore multiscale local spatial features. To verify the effectiveness of the multiscale setting on classification performance, we conduct ablation experiments. Based on the above parameter discussion results, we test the performance of single-scale input for each dataset. As shown in Fig. 3, three objective evaluation metrics—OA, AA, and Kappa—are adopted. N represents the single-scale input size with three options: λ, 2λ, and 3λ, and the corresponding classification performance is represented by blue, orange, and gray bars, respectively. Yellow bars reflect the multiscale input results. Without multiscale input mode, we only extract local spatial features at a single scale. After adding the multiscale input method, the information of different coverage areas can be used interactively. It can be concluded from Fig. 3 that for the three datasets, the three objective indicators have been greatly improved after adding the multiscale strategy. This shows that the local spatial features obtained by the multiscale aggregation method can obtain more discriminative local spatial features than the single-scale input. Especially for the Houston2013 dataset, due to the lower spatial resolution and higher scene complexity, the classification performance increases stepwise as N grows, which is in line with the results of parameter discussion. But the classification performance for multiscale input is even better, no matter which metric among OA, AA, and Kappa. Because for the Houston2013 data, the model not only needs a large coverage input but also expects more multiscale complementary information. Actually based on the experimental results, regardless of the dataset, the multiscale input has a positive contribution to the classification task.

2) Effect of the ViT: According to (9), we adopt the classical ViT module in GSFL to model the global spectral sequence properties. To verify the positive impact of ViT on the HSI and LiDAR joint classification task, we design ablation experiments. Fig. 4 depicts our experimental results based on three evaluation metrics (i.e., OA, AA, and Kappa) of three datasets. In Fig. 4, the blue bar represents the result without ViT, while the red bar represents the performance improvement after adding ViT. Without adding the ViT module, we only consider

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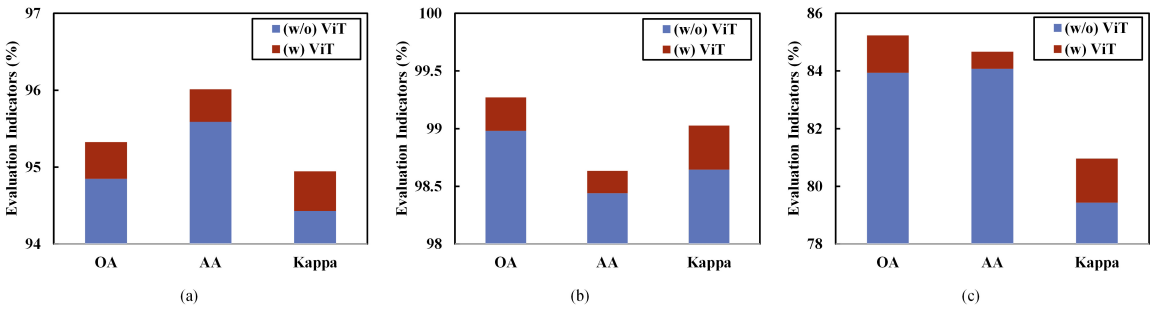


Fig. 4. Effects of the ViT module on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL.

TABLE I
COMPARISON OF THE CLASSIFICATION ACCURACIES (%) USING THE HOUSTON2013 DATA

No.	Class(Train/Test)	Performance									
		TB-CNN	HRWN	EndNet	MDL-RS	CoupledCNN	LSAF	FGCN	MAHiDFNet	MFT	GLT-Net
1	Healthy grass (20/1231)	90.01	84.97	91.36	94.24	94.24	91.08	88.29	82.01	**95.94**	94.09
2	Stressed grass (20/1234)	92.78	94.51	86.47	93.92	97.21	95.02	90.67	**97.94**	95.25	96.74
3	Synthetis grass (20/677)	**99.94**	99.63	96.75	98.85	99.65	**99.94**	96.66	**99.94**	99.61	99.57
4	Tree (20/1224)	97.76	95.55	88.69	97.87	98.53	73.44	91.56	96.00	**99.32**	98.15
5	Soil (20/1222)	97.52	97.62	91.19	96.11	99.56	99.39	95.83	96.19	**99.97**	99.80
6	Water (20/305)	91.41	94.75	93.15	91.93	98.23	97.57	91.67	96.82	**100**	97.08
7	Residential (20/1248)	88.13	89.94	59.07	88.54	94.88	74.80	86.13	91.83	**96.63**	94.94
8	Commercial (20/1224)	83.53	79.98	57.79	79.84	87.38	86.50	78.23	84.35	91.58	**91.59**
9	Road (20/1232)	80.80	76.54	66.94	85.71	85.43	82.22	77.70	84.10	90.37	**94.20**
10	Highway (20/1207)	79.99	75.42	69.15	87.46	**94.69**	88.48	81.98	78.61	94.09	91.92
11	Railway (20/1215)	84.57	90.83	64.05	84.52	**95.80**	90.58	79.29	87.58	85.29	93.18
12	Park lot 1 (20/1213)	75.89	84.34	43.31	76.17	85.56	85.76	72.98	69.78	88.87	**90.68**
13	Park lot 2 (20/449)	7.88	91.78	43.92	94.28	**98.60**	94.59	82.47	96.17	97.62	98.40
14	Tennis court (20/408)	99.41	**100**	97.55	98.09	**100**	99.36	96.03	**100**	**100**	99.90
15	Running track (20/640)	98.50	**100**	98.61	99.09	99.69	99.95	90.16	99.97	**100**	99.98
	OA (%)	86.22	88.80	74.44	89.92	94.35	88.70	85.53	88.90	94.71	**95.32**
	AA (%)	84.54	90.39	76.08	91.11	95.30	90.58	86.64	90.75	95.64	**96.01**
	Kappa (%)	85.07	87.90	72.87	89.10	93.89	87.78	84.38	88.01	94.28	**94.95**

the local spatial features F_{local} to perform the reconstruction task and calculate the final classification result. After adding ViT, we simultaneously consider the impact of local spatial features and global spectral dependencies on the classification performance. As shown in (10), we adaptively weight the local spatial prediction probability P_{local} and the global spectral prediction probability P_{global} to calculate the final fusion classification result. As can be seen from Fig. 4, with the addition of ViT, the three indicators have increased on the three datasets. For the Houston2013 data in particular, there are nearly equal-scale improvements across all the three metrics. Because the hyperspectral data of the Houston2013 dataset have a large number of spectral bands, the extraction of global spectral features is crucial and is reflected in all the metrics. Based on the experimental results, it can be concluded that ViT does make a positive contribution to the classification task.

E. Performance Comparison

To verify the advancement and effectiveness of the proposed GLT-Net model, we compare it with several state-of-the-art HSI and LiDAR joint classification models, including two-branch CNN (TBCNN) [27], hierarchical random walk network (HRWN) [36], EndNet [37], deep learning for the remote-sensing imagery classification (MDL-RS) [38], CoupledCNN [39], linear self-attention fusion (LSAF) [45], fractional Gabor convolutional network (FGCN) [40], multi-attentive hierarchical fusion net (MAHiDFNet) [46], and multimodal fusion transformer (MFT) [47]. Almost all of them adopt the deep CNN architecture, among which TBCNN, EndNet, MDL-RS, and FGCN are the feature-level fusion methods, HRWN and CoupledCNN use a decision fusion strategy, and LSAF, MAHiDFNet, and MFT introduce an attention mechanism. We optimize the parameters of all the compared methods on the same server as described in the original article. Also, for a fair comparison, absolutely identical training and testing samples are used, and the final experimental results are averaged over ten independent runs.

1) Comparison of Different Classification Methods: This article compares the GLT-Net method with four feature-level fusion methods (TBCNN, EndNet, MDL-RS, and FGCN), two decision fusion methods (HRWN and CoupledCNN), and three attention-based feature-level fusion methods (LSAF, MAHiDFNet, and MFT) from the perspective of visualization and quantitative analysis. Tables I–III illustrate the sample settings and objective classification results of different methods on three experimental datasets, where the best results in each row are marked in bold. Figs. 5–7 sequentially show hyperspectral false-color images, visualization of LiDAR data,

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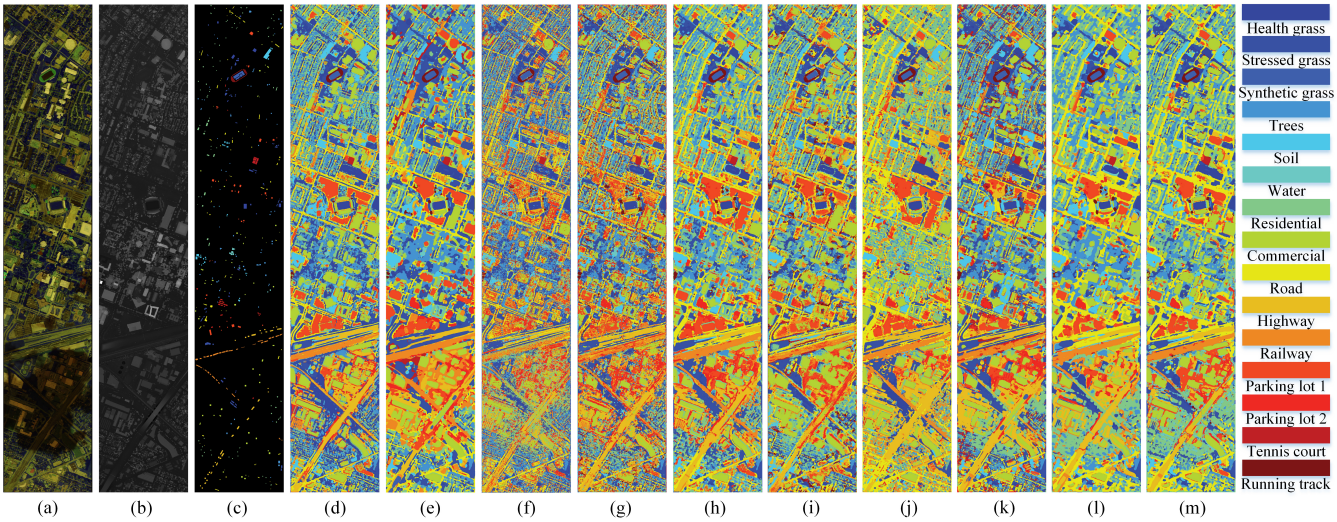


Fig. 5. Classification maps for the Houston2013 data obtained with different methods. (a) Pseudocolor image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (86.22%). (e) HRWN (88.80%). (f) EndNet (74.44%). (g) MDL-RS (89.92%). (h) CoupledCNN (94.35%). (i) LSAF (88.70%). (j) FGCN (85.53%). (k) MAHiDFNet (88.90%). (l) MFT (94.71%). (m) GLT-Net (95.32%).

TABLE II
COMPARISON OF THE CLASSIFICATION ACCURACIES (%) USING THE TRENTO DATA

No.	Class(Train/Test)	Performance									
		TB-CNN	HRWN	EndNet	MDL-RS	CoupledCNN	LSAF	FGCN	MAHiDFNet	MFT	GLT-Net
1	Apple trees (20/4014)	98.74	98.50	89.01	98.62	**99.65**	99.42	94.76	98.90	99.00	99.34
2	Buildings (20/2883)	93.42	96.93	92.04	95.48	96.60	91.25	93.48	91.15	94.11	**97.68**
3	Ground (20/459)	95.90	95.49	86.41	96.93	95.75	78.82	94.01	97.10	97.75	**97.89**
4	Woods (20/9103)	96.58	97.84	91.09	96.32	99.59	73.71	95.71	**99.97**	98.45	99.93
5	Vineyard (20/10481)	93.80	76.26	81.94	96.65	98.96	96.16	89.15	89.87	99.31	**99.80**
6	Roads (20/3154)	91.71	90.26	86.48	93.21	95.72	92.64	92.57	81.15	**99.53**	97.17
	OA (%)	95.08	89.49	87.16	96.35	98.62	88.70	92.73	93.45	98.51	**99.27**
	AA (%)	94.70	92.54	87.83	96.20	97.71	88.67	93.28	93.03	98.03	**98.63**
	Kappa (%)	93.80	86.06	83.21	95.15	98.17	85.25	90.35	91.28	98.01	**99.03**

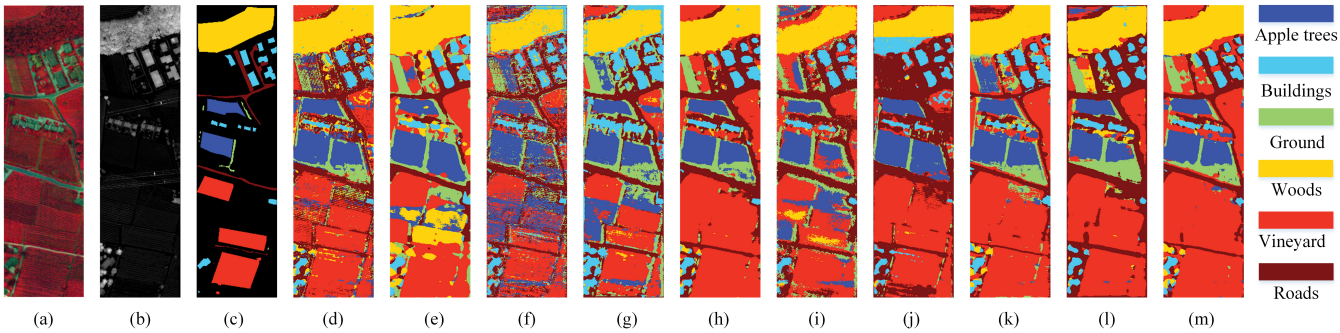


Fig. 6. Classification maps for the Trento data obtained with different methods. (a) Pseudocolor image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (95.08%). (e) HRWN (89.49%). (f) EndNet (87.16%). (g) MDL-RS (96.35%). (h) CoupledCNN (98.62%). (i) LSAF (88.70%). (j) FGCN (92.73%). (k) MAHiDFNet (93.45%). (l) MFT (98.5 1%). (m) GLT-Net (99.27%).

ground-truth plots, and classification maps for different methods on three experimental datasets, and the colors used for each category are also stated.

For the Houston2013 dataset, which contains 15 types of ground objects, some areas are occluded by clouds and fog, resulting in high scene complexity. We can draw the following conclusions from Table I. First, for the pure feature-level fusion CNN frameworks, multiple types of feature extraction outperform single-type feature extraction. For example, for HSI and LiDAR data, TB-CNN extracts spectral features and spatial features, respectively, and fuses them, which achieves higher accuracy than the fusion methods that only extract a

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TABLE III
COMPARISON OF THE CLASSIFICATION ACCURACIES (%) USING THE MUUFL DATA

No.	Class(Train/Test)	Performance									
		TB-CNN	HRWN	EndNet	MDL-RS	CoupledCNN	LSAF	FGCN	MAHiDFNet	MFT	GLT-Net
1	Trees (20/23226)	84.52	86.36	79.70	85.39	89.12	58.44	83.44	79.06	**89.48**	88.52
2	Mostly grass (20/4250)	**79.64**	74.15	74.33	67.83	73.93	54.53	67.57	67.76	63.73	79.05
3	Mixed ground surface (20/6862)	65.23	63.64	61.31	54.20	58.47	51.50	53.25	64.55	61.55	**68.23**
4	Dirt and sand (20/1806)	80.66	55.98	74.11	77.30	86.36	73.70	78.56	83.65	82.67	**90.42**
5	Road (20/6667)	82.04	62.70	83.83	71.41	80.46	60.26	63.80	**87.81**	84.78	87.21
6	Water (20/446)	97.76	98.92	96.57	98.45	99.53	99.60	90.34	45.13	**99.78**	98.39
7	Building shadow (20/2213)	81.62	83.63	86.16	86.59	82.42	71.26	82.65	**92.36**	80.60	88.66
8	Building (20/6220)	94.23	94.71	90.33	89.55	**95.39**	83.19	88.74	82.01	92.36	95.29
9	Sidewalk (20/1365)	49.60	36.15	60.35	61.21	63.32	44.73	51.25	**74.97**	43.44	60.74
10	Yellow curb (20/163)	70.31	51.29	**91.17**	72.15	82.15	71.66	68.16	81.10	42.33	79.14
11	Cloth panels (20/249)	95.58	91.12	95.38	95.58	95.14	**98.35**	95.22	98.07	95.98	95.74
	OA (%)	81.45	78.09	78.50	78.01	82.69	61.61	75.52	78.15	81.80	**85.23**
	AA (%)	80.11	72.61	81.20	78.15	82.39	69.75	74.82	77.86	76.06	**84.67**
	Kappa (%)	76.27	71.80	72.78	71.90	77.70	53.38	68.97	72.45	76.37	**80.96**

single type of deep features (such as EndNet and FGCN). This illustrates the importance of more targeted feature extraction. Second, the sufficient interaction of multimodal information can improve the classification performance, such as the cross-fusion method designed by MDL-RS. Third, the attention mechanism has a positive effect on feature-level fusion, such as LSAF, MAHiDFNet, and MFT perform better than most pure feature-level fusion methods. Fourth, a suitable decision-level fusion strategy helps further improve the classification performance. For example, HRWN performs better than pure feature-level fusion methods except MDL-RS, and CoupledCNN achieves the third best results. In contrast, our GLT-Net method achieves the best performance on OA, AA, and Kappa on the Houston2013 dataset and achieves the highest classification accuracy for five of the 15 categories. We combine the above advantages in an end-to-end framework, not only considering the global–local features and spatial–spectral features but also applying the attention mechanism and adding decision-level fusion strategies on the basis of feature-level fusion, which can effectively improve the classification results. As can be seen from Table I, for the Houston2013 dataset, we achieve about 0.5% improvement on all the three metrics compared with the suboptimal approach. From the classification map in Fig. 5, it can be seen that our GLT-Net method has clear edges for the classification results of residential and commercial areas, especially in the areas occluded by clouds, which is consistent with Table I. This indicates that GLT-Net can better achieve the fusion of the spatial–spectral information of HSI and the elevation information of LiDAR data. Specifically, in MLSFL, under the cooperation of multiscale aggregated CNN and SA-GDR module, edge information that is easily ignored in single-scale local spatial patches is fully learned. Second, in GSFL, modeling the global spectral dependence enables our method to obtain more discriminative features. Finally, in GLFFC, multiscale and multimodal global and local information are fused to further improve classification. For the Trento and MUUFL datasets with higher spatial resolution and fewer feature categories, the performance of different classification methods also shows a similar trend.

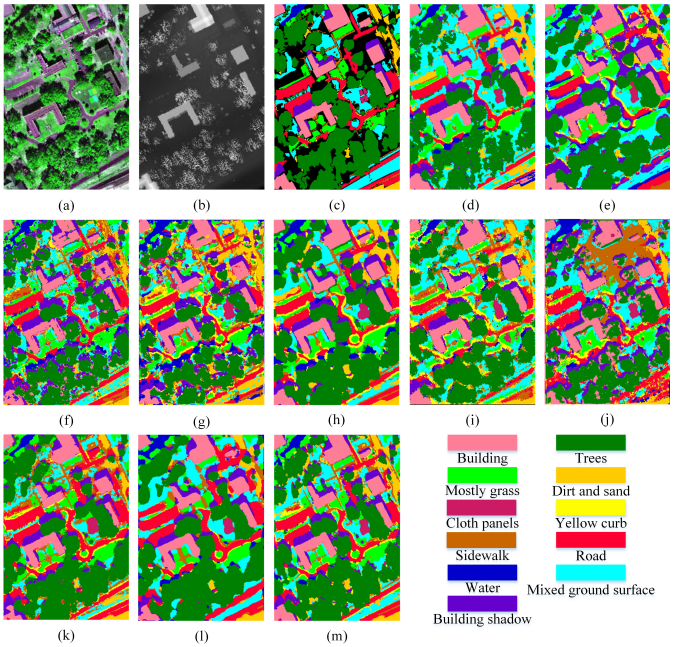


Fig. 7. Classification maps for the MUUFL data obtained with different methods. (a) Pseudocolor image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (81.45%). (e) HRWN (78.09%). (f) EndNet (78.50%). (g) MDL-RS (78.01%). (h) CoupledCNN (82.69%). (i) LSAF (61.61%). (j) FGCN (75.52%). (k) MAHiDFNet (78.15%). (l) MFT (81.80%). (m) GLT-Net (85.23%).

Table II lists the quantitative classification results on the Trento data. GLT-Net achieves the best performance on three objective indicators, which is more than 0.5% higher than the suboptimal model, as well as the highest level in four of the six land object categories. As shown in Table III, our method also achieves the highest OA, AA, and Kappa on the MUUFL data, with an improvement of around 3%, and achieves the highest accuracy on two ground objects. Figs. 6 and 7 shows plot of the visual classification results for the Trento and

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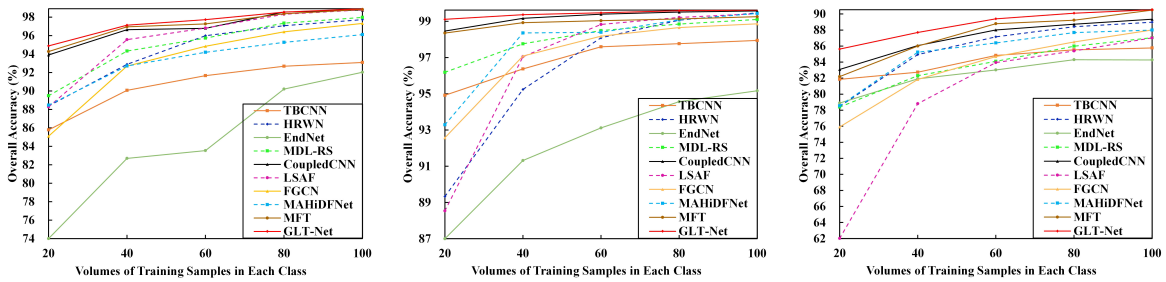


Fig. 8. Classification performance with different volumes of training samples using: (a) Houston2013, (b) Trento, and (c) MUUFL.

TABLE IV
COMPARISON OF MODEL COMPLEXITY AND TRAINING TIME (S) OF DIFFERENT MODELS ON THE HOUSTON2013 DATA

Methods	TB-CNN	HRWN	EndNet	MDL-RS	CoupledCNN	LSAF	FGCN	MAHiDFNet	MFT	GLT-Net
Trainable Params	1.72M	1.37M	90.48K	120.37K	102.33K	2.06M	1.04M	74.13M	305.11K	545.63K
Computation Cost(MFLOPs)	1.80	1.44	0.56	0.83	2.28	117.38	13.09	155.43	50.32	312.28
Training time(s)	350.75	150.34	90.06	363.46	248.56	342.32	11511.56	9009.14	4278.72	562.6

MUUFL datasets. The visualization of the classification results of our GLT-Net method is closer to the ground-truth map. For the Trento data, our classification results are smoother with fewer isolated class points due to the stronger information fusion ability, which is consistent with the actual situation. For example, for the fifth category (Vineyard) with the largest number of samples, the continuity of the visualization results of our proposed method is the best, which is in line with the characteristics of ground objects. On the MUUFL data, our method fully exploits the global spectral dependence and has better discriminative ability for mixed ground surface. In conclusion, the effectiveness and state-of-the-art of our GLT-Net model are validated in both objective and visual comparisons.

2) Performance With Different Training Sample Volumes: To further verify the robustness and effectiveness of our proposed GLT-Net model, we set the number of training samples stepwise and study the changing trend of the overall classification accuracy. Specifically, we randomly select a fixed number of samples from each category for training and the rest for testing. The number of picks is increased from 20 to 100 in steps of 20. All the classification methods have the same sample situation, and finally take the average of ten independent repeated experiments. Fig. 8 shows the classification performance of the proposed GLT-Net model and other comparative methods under different volumes of training samples, described in terms of OA. It can be seen from Fig. 8 that the classification accuracy of all the methods increases with the increase in the number of training samples, which reflects the positive impact of a larger number of training samples on the DL-based classification methods. More importantly, our GLT-Net always achieves the best performance when the number of training samples is same. Especially when the number of training samples is small (e.g., 20), our model has more obvious advantages. This benefits from the combined utilization of the local spatial features and global spectral dependencies of HSI and LiDAR data and the sufficient interaction of multimodal information.

3) Computational Complexity Analysis: Table IV shows the model complexity of different networks, including the number of trainable weight parameters updated during back-propagation and computational cost. It can be found that the number of parameters of the proposed model is in the middle level, which is the fourth least among the ten models. The EndNet method has the least number of trainable parameters, while MAHiDFNet uses the largest number of parameters. We compute the computational cost by million floating-point operations (MFLOPs). It can be observed that the proposed method has the highest MFLOPs value because we adopt a multiscale input strategy for multimodal data, which increases the computational cost while improving the classification accuracy. In addition, Table IV also shows the training time of different methods on the Houston2013 dataset. The training time of our proposed method is comparable to other state-of-the-art fast algorithms (in the ten square level), and more importantly, within the acceptable training time range, GLT-Net can achieve the best classification results.

IV. CONCLUSION

Considering the combined utilization of local and global spatial–spectral information of HSI and LiDAR data and the efficient interaction of multimodal data, this article proposes an end-to-end DL model called GLT-Net. Specifically, we use the powerful local spatial context extraction ability of CNN and the advantages of transformers in global relation modeling and combine with multiscale feature fusion and decision fusion for joint classification. We designed three key steps: MLSFL, GSFL, and GLFFC. First, multiscale local spatial features are extracted in MLSFL, and multimodal information is fused at the feature level. Second, GSFL models the global spectral context properties of fused features from the perspective of spectral sequences. Finally, the global–local features are decision-fused in GLFFC for better classification results.

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Under the constraint of the joint loss, the feature learning process and the classification process promote each other. To demonstrate the effectiveness of the proposed GLT-Net, we conduct rich experiments on three popular HSI and LiDAR datasets. Through quantitative and visual analysis, we can demonstrate that the proposed method outperforms other state-of-the-art methods.

In addition, GLT-Net still has limitation that needs to be further explored. For example, we need a more detailed study on the problem of weight assignment in multitask mode. Therefore, one of the main research contents in the future is to develop an efficient adaptive weight assignment scheme. We will further explore how to modify the transformer model to study the local–global correlation of spectra. In addition, how to alleviate the dependence of the DL-based models on the number of samples is worth exploring. Another future work focus is semi-supervised classification of HSI and LiDAR with unlabeled samples.

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Kexing Ding received the B.S. degree in measurement and control technology and instrument from Liaoning University, Liaoning, China, in 2020. He is currently pursuing the Ph.D. degree in control science and engineering with Hunan University, Changsha, China.

His research interests focus on deep learning and joint processing of HSI and LiDAR data.



Ting Lu (Member, IEEE) received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2011 and 2017, respectively.

From 2014 to 2015, she was a Visiting Ph.D. Student with the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, supported by the China Scholarship Council. In 2017, she became an Assistant Professor with Hunan University, where she is currently an Associate Professor with the College of Electrical and Information Engineering. Her research interests include image processing, pattern recognition, and applications in remote sensing.



Wei Fu received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2012 and 2019, respectively.

From 2016 to 2017, he was a Visiting Ph.D. Student with the Department of Electrical and Computer Engineering, University of Iceland, Reykjavik, Iceland, supported by the China Scholarship Council. Since 2019, he has been an Assistant Professor with the College of Information Science and Engineering, Hunan University. His research interests focus on remote sensing image processing and analysis.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May to October 2001, he was a Research Associate with the Department of Computer Science, Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K., working with Prof. John Shawe-Taylor. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, Hong Kong University of Science and Technology. In 2013, he was granted the National Science Fund for Distinguished Young Scholars in China. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He is also a Chang-Jiang Scholar Professor appointed by the Ministry of Education of China. He has authored or coauthored more than 200 refereed articles. His professional interests include compressive sensing, sparse representation, image processing, and pattern recognition.

Dr. Li is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and a member of the Editorial Board of the *Information Fusion* and the *Sensing and Imaging*. He was a recipient of two Second-Grade National Awards at the Science and Technology Progress of China in 2004 and 2006.



Fuyan Ma received the B.S. degree from the College of Electronics and Information Engineering, South Central University for Nationalities, Wuhan, China, in 2018. He is currently pursuing the Ph.D. degree with the College of Electrical and Information Engineering, Hunan University, Changsha, China.

His research interests include image processing, affective computing, and multimodal information fusion.

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